METHOD 05 / A COMPLETE BEGINNER GUIDE

# Rolling correlation

**Question this method answers:** Right now, using only the last W matched daily returns, how strongly do X and Y co-move?

W is an observation count (e.g. 20 trading days), not a calendar-month label. Short windows are noisy; longer windows smooth but lag regime shifts.

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Prepared for: 05 - Rolling Correlation. Experiments v001 is complete on snapshot 2026-11-03_majors_alpari_D1.

02 / THE BASIC IDEA

# What is rho_t^(W)?

```
rho_t^(W) = Corr( last W observations of x, last W of y )
```

At each end date t, take the previous W rows of both return series and compute Pearson r. Move t forward one day to get a new value.

## Is this number a percent?

| Mistake | Truth |
| --- | --- |
| "r_20 is 20% something" | No. 20 is the window length in observations; r is still between -1 and +1. |
| "High r_20 means 20 winning days" | No. It describes co-movement of returns inside the window. |

03 / RETURNS BEFORE ROLLING

# FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

v001 used daily log returns of close, quote unchanged. Example: 1.1000→1.1110 → about +1.0000% simple, ~0.9950% log.

Reverse quote flips log sign. Missing ≠ 0. Four prices → three returns.

Rolling correlation is computed on these return series, not on prices. Mixing return types between baseline and rolling breaks comparability with method 01 outputs.

04 / DATA CHOICE

# Which series this method uses

Rolling Pearson on **matched daily log returns** (same as method 01 baseline in this repo). Each window is W consecutive trading days present in both series after merge.

Windows used in v001: $W \in \{20, 60, 100\}$.

The unfinished last D1 bar was dropped in v001 after a tick-volume check. Your rolling file should use the same complete-bar rule so r_20 lines up with the snapshot metadata.

Pair names like AUDUSD-NZDUSD mean two return columns merged on date — not a cross-rate formula.

# Research notes (section 05)

Section 05 in forex-formulas-explained.pdf defines rolling correlation as local Pearson r over a moving window.

Dynamic conditional correlation models (Engle, 2002) popularized time-varying correlation for risk management. Rolling r is a transparent, non-model shortcut — useful for monitoring even when you do not fit full DCC.

# Symbol dictionary

| Symbol | Meaning |
|---|---|
| W | Window length in observations (not calendar days unless data is daily and complete). |
| rho_t^(W) | Rolling correlation at end date t. |
| r_20, r_60, r_100 | Latest rolling values for each window in v001 outputs. |
| baseline_r | Full-sample Pearson r from method 01 v001. |

07 / WINDOW MECHANICS

# Formula explained

For each t, slice x[t-W+1:t] and y[t-W+1:t] (inclusive). Require exactly W valid pairs; gaps in data shrink effective history unless you impute (not done in v001).

Early dates before W rows exist are skipped or marked missing in output CSVs.

Each window recomputes means from scratch inside the slice — a big move ages out only when it leaves the W-day tail. That is why r_20 reacts faster than r_100 on the same pair.

Calendar holidays remove rows from the count; W = 20 means twenty observed returns, not twenty calendar days on a wall calendar.

08 / W = 5

# Teaching hand example (separate from Alpari)

Made-up decimal returns (last five rows drive the score):

| Day | x | y |
|---|---|---|
| 4 | 0.004 | 0.003 |
| 5 | -0.001 | -0.002 |
| 6 | 0.003 | 0.002 |
| 7 | 0.0 | 0.001 |
| 8 | 0.002 | 0.001 |

```
rho_roll (W=5 on full 8-day teach set uses last 5) = 0.902194
```

This toy series is only for arithmetic and Python checks. Alpari v001 pair CSVs contain thousands of rolling rows — never mix those filenames with this teaching table.

Recompute Pearson on the last W rows only; including earlier rows would answer a different window question.

09 / ALPARI D1 SNAPSHOT

# Real results — experiments/v001

Source: experiments/v001/NOTES.md and outputs/WATCHLIST.md (only v001). Series: daily log return of close. Statistic: rolling Pearson r with W = 20, 60, 100.

## Short watchlist (latest rolling vs baseline)

| Pair | baseline | r_20 | r_60 | r_100 |
|---|---|---|---|---|
| AUDUSD-NZDUSD | 0.829 | 0.263 | 0.607 | 0.706 |
| EURUSD-USDCHF | -0.749 | -0.729 | -0.692 | -0.755 |
| AUDUSD-USDCAD | -0.647 | -0.302 | -0.526 | -0.523 |
| EURUSD-GBPUSD | 0.642 | 0.925 | 0.875 | 0.851 |
| NZDUSD-USDCAD | -0.588 | -0.398 | -0.531 | -0.564 |

WATCHLIST highlights: AUDUSD-NZDUSD baseline 0.829 vs r_20 0.263 (large gap); EURUSD-GBPUSD r_20 0.925 above baseline 0.642. Short windows can diverge sharply from full history.

# More v001 tiers (summary)

Secondary examples: EURUSD-AUDUSD r_20 ≈ -0.004 vs baseline 0.556; GBPUSD-USDCHF r_20 ≈ -0.728 vs baseline -0.486.

Low-value pairs often stay near zero on r_100 (e.g. USDCAD-EURGBP r_100 ≈ 0.007) — WATCHLIST says safe to ignore unless r_20 spikes.

Monitoring order from WATCHLIST: refresh r_20 first each D1 close, confirm with r_60 and r_100, escalate on sign flips or large baseline gaps.

# Daily majors workflow

1) Fix snapshot and merge rules. 2) Compute baseline r once. 3) Append rolling r_20/60/100 each day. 4) Compare to baseline — not to arbitrary thresholds from other pairs.

std20 and std100 in NOTES measure rolling-series volatility; high std20 means r_20 is jumping — treat short windows as noisy.

# Limits

Rolling r does not forecast tomorrow, does not pick hedge size, and is not a percent. Changing W changes the answer — always label W.

Independence assumptions for classic p-values fail on daily FX; use rolling r as descriptive monitoring, as in v001.

A single spike in r_20 can be noise — WATCHLIST compares baseline gaps for a reason. Do not trade on one window alone without charts and baseline context.

# Software check

```
W = 5
x = [0.01, -0.005, 0.002, 0.004, -0.001, 0.003, 0.0, 0.002]
y = [0.008, -0.004, 0.001, 0.003, -0.002, 0.002, 0.001, 0.001]
rho = pearsonr(x[-W:], y[-W:])[0]
```

Expected teaching rho (last W=5) = 0.902194. v001 pair numbers come from run_rolling.py outputs, not from this toy list.

## 14 / 20 VS 60 VS 100

# Window choice

W=20 reacts fast but noisy (see AUDUSD-NZDUSD). W=100 tracks baseline more closely for stable pairs like EURUSD-USDCHF. Use multiple W as a triangulation, not a single alarm.

std20 in v001 NOTES flags pairs whose r_20 series jumps — pair high std20 with charts before calling a regime break.

If r_20, r_60, and r_100 disagree in sign, treat the pair as unstable until you understand the recent window.

15 / QUICK QUESTIONS

# Practice

**Q:** W=20 on daily data means 20 calendar days? **A:** Twenty observations — usually trading days if your series has no gaps.

**Q:** Can r_20 exceed +1? **A:** No.

**Q:** baseline_r vs r_100 — should they match? **A:** Not necessarily; baseline uses full history while r_100 is only the latest 100-day window.

**Q:** Where do Alpari numbers live? **A:** experiments/v001/NOTES.md and outputs/WATCHLIST.md — not in the teaching hand example on page 8.

Monitoring habit: note sign of r_20 and whether it agrees with r_60 and r_100 before escalating a pair on the watchlist.

# Sources

**[1] Engle, R. (2002). Dynamic Conditional Correlation. Journal of Business & Economic Statistics, 20(3), 339–350.**
Time-varying correlation context for rolling monitoring.
Open source

**[2] SciPy documentation: scipy.stats.pearsonr.**
Pearson r inside each window.
Open source

**[3] forex-formulas-explained.pdf — section 05; experiments/v001/NOTES.md; outputs/WATCHLIST.md.**
Repo formulas and frozen v001 numbers.
Open source